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JEE MAIN CRASH COURSE

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- LATEST NTA PATTERN
- CONCEPT VIDEOS
- LIVE DOUBT SESSIONS
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Designed by
Math Maestro

Anup Sir

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JEE Mains 2020 Jan Chapter wise Question Bank

Functions

Q1

If $g(x) = x^2 + x - 1$ and $g(f(x)) = 4x^2 - 10x + 5$, then find $f\left(\frac{5}{4}\right)$.

यदि $g(x) = x^2 + x - 1$ और $g(f(x)) = 4x^2 - 10x + 5$, तब $f\left(\frac{5}{4}\right)$ का मान है–

(1) $\frac{1}{2}$ (2) $-\frac{1}{2}$ (3) $-\frac{1}{3}$ (4) $\frac{1}{3}$

7th Jan Morning

Sol

(2)

$g(f(x)) = f^2(x) + f(x) - 1$

$g\left(f\left(\frac{5}{4}\right)\right) = 4\left(\frac{5}{4}\right)^2 - 10.\ \frac{5}{4} + 5 = -\frac{5}{4}$

$g\left(f\left(\frac{5}{4}\right)\right) = f^2\left(\frac{5}{4}\right) + f\left(\frac{5}{4}\right) - 1$

$-\frac{5}{4} = f^2\left(\frac{5}{4}\right) + f\left(\frac{5}{4}\right) - 1$

$f^2\left(\frac{5}{4}\right) + f\left(\frac{5}{4}\right) + \frac{1}{4} = 0$

$\left(f\left(\frac{5}{4}\right) + \frac{1}{2}\right)^2 = 0$

$f\left(\frac{5}{4}\right) = -\frac{1}{2}$

02

Let $f(x) = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}}$ then inverse of f(x) is

माना $f(x) = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}}$ तब f(x) का प्रतिलोम फलन है :

(1) $\frac{1}{4}\log_8\left(\frac{1+x}{1-x}\right)$ (2) $\frac{1}{2}\log_8\left(\frac{1-x}{1+x}\right)$ (3) $\frac{1}{4}\log_8\left(\frac{1-x}{1+x}\right)$ (4) $\frac{1}{2}\log_8\left(\frac{1+x}{1-x}\right)$

8^{th} Jan Morning

Sol

(1)

$$y = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}}$$

$$\frac{1+y}{1-y} = \frac{8^{2x}}{8^{-2x}}$$

03

Let $f(x) = \frac{x[x]}{x^2+1} : (1, 3) \to R$ then range of f(x) is (where [.] denotes greatest integer function)

माना $f(x) = \frac{x[x]}{x^2+1} : (1, 3) \to R$ तो f(x) का परिसर है (जहाँ [.] महत्तम पूर्णांक फलन है)

(1) $\left(0, \frac{1}{2}\right) \cup \left(\frac{3}{5}, \frac{7}{5}\right]$ (2) $\left(\frac{2}{5}, \frac{1}{2}\right) \cup \left(\frac{3}{5}, \frac{4}{5}\right]$

(3) $\left(\frac{2}{5}, 1\right) \cup \left(1, \frac{4}{5}\right]$ (4) $\left(0, \frac{1}{3}\right) \cup \left(\frac{2}{5}, \frac{4}{5}\right]$

8^{th} Jan Evening

Sol

(2)

$$f(x) \begin{cases} \frac{x}{x^2+1}; & x \in (1,2) \\ \frac{2x}{x^2+1}; & x \in [2,3) \end{cases}$$

$\therefore$ f(x) is a decreasing function

$\therefore$ f(x) ह्रासमान फलन है।

$\therefore$ $y \in \left(\frac{2}{5}, \frac{1}{2}\right) \cup \left(\frac{6}{10}, \frac{4}{5}\right]$

$\Rightarrow$ $y \in \left(\frac{2}{5}, \frac{1}{2}\right) \cup \left(\frac{3}{5}, \frac{4}{5}\right]$

04

Find the number of solution of $\log_{1/2} |\sin x| = 2 - \log_{1/2} |\cos x|$, $x \in [0,2\pi]$

$x \in [0,2\pi]$ में $\log_{1/2} |\sin x| = 2 - \log_{1/2} |\cos x|$ के हलों की संख्या है–

(1) 2 (2) 4 (3) 6 (4) 8

9^{th} Jan Morning

Sol

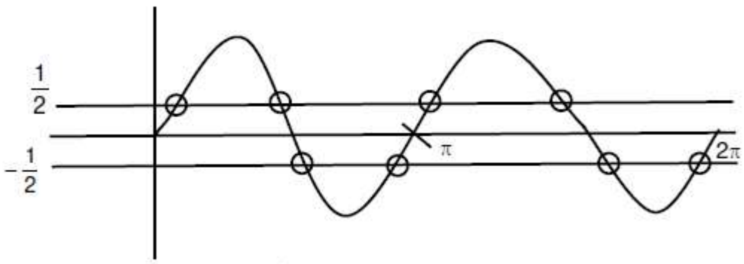
(4)

$\log_{1/2}|\sin x| = 2 - \log_{1/2}|\cos x|$

$\log_{1/2}|\sin x \cos x| = 2$

$|\sin x \cos x| = \frac{1}{4}$

$\sin 2x = \pm\frac{1}{2}$



Number of solution हलों की संख्या = 8.

05

if f(x) and g(x) are continuous functions, fog is identity function, $g'(b) = 5$ and $g(b) = a$ then $f'(a)$ is
यदि f(x) तथा g(x) सतत् फलन है, fog तत्समक फलन है, $g'(b) = 5$ तथा $g(b) = a$ तब $f'(a)$ है–

(1) $\frac{2}{5}$ (2) $\frac{1}{5}$ (3) $\frac{3}{5}$ (4) 5

9th Jan Evening

Sol

(2)

$f(g(x)) = x$

$\Rightarrow f'(g(x)).\ g'(x) = 1$

Put $x = b$

$\Rightarrow f'(g(b))\ g'(b) = 1$

$\Rightarrow f'(a) \times 5 = 1$

$\Rightarrow f'(a) = \frac{1}{5}$

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